

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain architecture of DBMS with a neat and clean diagram. (CO2)
- Q.24 Explain various normal forms based on functional dependencies with suitable example. (CO3)
- Q.25 What do you mean by locking in concurrency control? Explain two-phase locking in detail. (CO5)

No. of Printed Pages : 4
Roll No.

223853

5th Sem/ Artificial Intelligence & Machine Learning Subject : Relation Database Management System

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is a relational database management system? (CO1)
- a) MongoDB b) SQLite
c) Redis d) Cassandra
- Q.2 In a database, what is the purpose of the "JOIN" operation? (CO3)
- a) To insert new record
b) To update existing record
c) To retrieve data from multiple records
d) To delete records
- Q.3 What is a relation in RDBMS? (CO1)
- a) Key b) Table
c) Row d) Data Types

- Q.4 Normalization helps in _____ (CO2)
- Avoiding redundancy
 - Increasing performance
 - Reducing query time
 - Reducing storage space
- Q.5 What does SQL stand for? (CO4)
- Structured Query Language
 - Simple Query Language
 - Sequential Query Language
 - None of the above
- Q.6 Which normalization form eliminates transitive dependency? (CO2)
- 1NF
 - 2NF
 - 3NF
 - BCNF

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Name two workers behind the scene in DBMS. (CO1)
- Q.8 Define a data model. (CO1)
- Q.9 What do you mean by data redundancy? (CO1)

(2) 223853

- Q.10 Define an Entity Type. (CO2)
- Q.11 Write SQL commands to delete a record from table. (CO4)
- Q.12 What is the use of Revoke command? (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What are the advantages of using database approach? (CO1)
- Q.14 What do you mean by database users? Explain its types. (CO1)
- Q.15 Define different types of constraints used in RDBMS. (CO2)
- Q.16 Explain in brief various DBMS interfaces. (CO2)
- Q.17 What are the properties of relation in RDBMS? (CO5)
- Q.18 Explain various types of keys used in RDBMS. (CO3)
- Q.19 What do you mean by non-loss decomposition? (CO3)
- Q.20 What do you mean by normalization? What is its need? (CO3)
- Q.21 Explain various forms of SELECT command with suitable example. (CO4)
- Q.22 Explain some challenges of MySQL? (CO4)

(3) 223853